EX.NO: 7

17-08-2026

DEVELOP A PYTHON SCRIPT USING NMAP TO IDENTIFY ACTIVE HOSTS IN A NETWORK

AIM:

To develop a python script using Nmap to identify active hosts in a network.

ALGORITHM:

STEP 1: Start the program & Download and install Nmap & Npcap.

STEP 2: Import the required modules, os and nmap.

STEP 3: Add the Npcap DLL directory (C:\Windows\System32\Npcap) to the system DLL search path.

STEP 4: Create an Nmap PortScanner object and specify the path to the Nmap executable.

STEP 5: Read the target network address (e.g., 192.168.1.0/24) from the user.

STEP 6: Perform a host discovery scan on the specified network using the Nmap -sn option**.**

STEP 7: Retrieve the list of hosts detected by the scanner.

STEP 8: Iterate through each detected host.

STEP 9: Retrieve the current state of each host, such as up or down.

STEP 10: Display the IP address and corresponding host state on the screen.

STEP 11: Stop the program.

CODING

```
import os, nmap

os.add_dll_directory(r"C:\Windows\System32\Npcap")

scanner = nmap.PortScanner(nmap_search_path=(r"C:\Program Files
(x86)\Nmap\nmap.exe",))

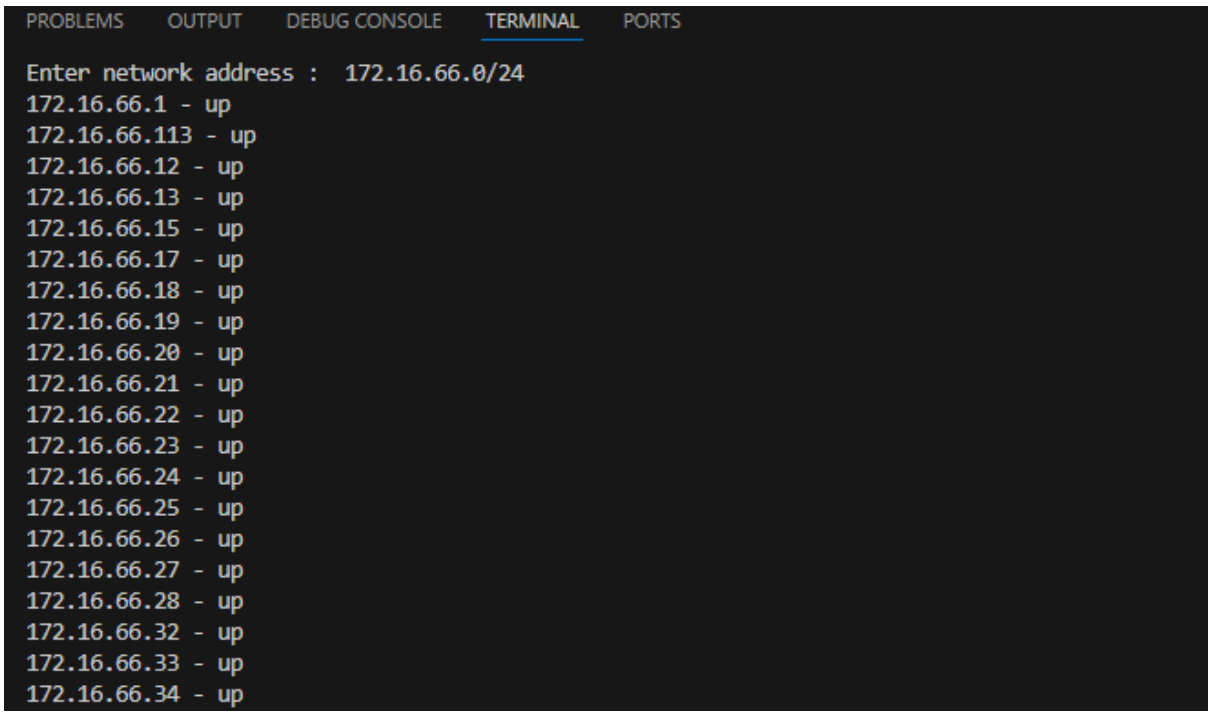

target = input("Enter network address : ")

scanner.scan(hosts=target, arguments="-sn")


for host in scanner.all_hosts():

    print(host, "-", scanner[host].state())
```

OUTPUT



RESULT

Thus, the program to develop a python script using Nmap to identify active hosts in a network is executed successfully.